RAS AL KHAIMAH GOVERNMENT · DEPARTMENT OF FINANCE

GovBudget — Test Results and Business Objective Validation

Evidence of the application tested against the business objectives it was built to meet, prepared for EGA / IDO evaluation ahead of sandbox provisioning.

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Environment tested — current hosted instance, budget year 2026, live departmental data

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1. Purpose and scope

This document reports what has been tested in GovBudget and what the results were, framed against the business objectives the system exists to serve. It is intended to support EGA's evaluation while the Azure sandbox is being provisioned.

It deliberately reports both what passes and what has **not** yet been exercised. Two objectives are not yet fully evidenced, and they are stated as plainly as the ones that pass — an evaluation is only useful if the gaps are visible.

2. How these results were produced

Three kinds of verification were used, and each result below states which applies:

Method	What it establishes
Data verification	Direct queries against the live database, checking that the figures the system produces actually reconcile. This is the strongest evidence: it tests outcomes, not intentions.
Application testing	Functions exercised through the application in a browser, signed in as a user with the relevant role.
Code verification	Inspection of the implementation where behaviour cannot be observed from data alone — for example password hashing parameters or outbound network calls.

> **Scale of the data behind these results**
> These are not results from a demonstration data set. They come from the department's working 2026 budget.

Budget lines, FY2026	Budget value	Employees costed	Programmes / activities
1,034	335.0 M AED	470	14 / 37

Audit records
626

3. Results summary

#	Business objective	Result	Basis
1	Shift from input-based to performance-based budgeting: cost attributed to programmes and activities, not just accounts	Passed	Data verification
2	Complete attribution of manpower cost to the activities staff actually work on	Passed	Data verification
3	Overhead redistribution: support programme cost carried onto the mandate work it enables, without altering the underlying budget	Passed	Data verification
4	Unit costing: express cost per hour of staff time as a basis for costing services and outputs	Passed	Data verification
5	Performance layer: KPIs, activity outputs and maturity captured alongside cost	Partial	Data verification
6	Governed budget cycle: prepare, submit, review, return, approve — with an evidence trail	Not yet exercised	Code verification only
7	Management reporting sufficient to support budget decisions	Passed	Application testing
8	Access control and auditability appropriate to government financial data	Passed	Code verification + data
9	Operable inside a restricted environment with no outbound internet	Passed	Code verification

4. Detailed results

Objective 1 — Cost attributed to programmes and activities Passed

The FY2026 budget is fully structured against the performance-based model rather than against accounts alone.

Entity	Budget lines	Budget (AED)
RCUD — Customs Department	883	324,574,511
RDAM — Antiquities and Museums	148	8,925,708
RDOF — Department of Finance	3	1,525,000

Every line carries an activity and a general ledger account, and each activity belongs to a programme classified as either Mandate or Support. That structure is what makes the reporting in objectives 3, 4 and 7 possible. Figures are rounded to whole dirhams.

Objective 2 — Complete manpower attribution Passed

This is the objective most likely to fail quietly in a costing system: salary that is budgeted but never attached to any activity disappears from programme reporting while still being spent.

Check	Result
Employees costed for FY2026	470
Total annual manpower cost	87,363,024 AED
Total allocated to activities	87,363,024 AED
Allocation rows	876
Employees not fully allocated	**0**

Attribution is 100%. Allocated cost equals budgeted cost to the dirham, and no employee is partially or wholly unattached to activities. The application enforces this: it refuses an allocation exceeding an employee's annual cost, and reports anyone under-allocated on a dedicated variance screen.

Objective 3 — Overhead redistribution Passed

Support programme cost is redistributed onto the mandate programmes it enables, so that programme cost can be reported fully loaded rather than only as direct cost.

One allocation run is posted. It moves the entire direct cost of the support programme:

Direction	Programme	Amount (AED)
Out	DAM-07 — Governance & Corporate Support (Support)	8,685,800.12
In	DAM-04 — Archaeological Sites & Collections	3,985,249.48
In	DAM-05 — Archaeological Excavation & Research	2,452,461.21
In	DAM-03 — Museum Spaces, Content and Exhibitions	1,532,788.25
In	DAM-02 — Community Awareness of Culture and Heritage	408,743.54
In	DAM-01 — Intangible Heritage Documentation	204,371.77
In	DAM-06 — Partnerships & Knowledge Exchange	102,185.87
Total in		**8,685,800.12**

Amount out equals amount in exactly, and that figure is independently confirmed as DAM-07's full direct cost (4,897,577.93 in budget lines plus 3,788,222.19 in allocated manpower). Total entity cost is therefore unchanged by allocation: cost moves, it is neither created nor lost.

Separately confirmed: the run writes only to its own tables. Budget lines and manpower allocations are not modified, so the Direct basis always continues to show the budget exactly as entered. Superseded runs are retained rather than overwritten.

Objective 4 — Unit costing (cost per hour) Passed

Check	Result
Employees with a calculated hourly rate	470 of 470
Blended standard rate (excluding vacant posts)	104.51 AED / hour
Vacant budgeted posts, correctly excluded	15

The rate divides annual cost by *productive* hours — contracted hours less paid public holidays and annual leave — rather than by contracted hours. Because paid leave already sits inside the annual salary, dividing by contracted hours would recover only 86.15% of it (1,792 productive hours against 2,080 contracted) and under-cost every activity by 13.85%. Dividing by productive hours means hours multiplied by rate reconciles back to total salary cost, preserving the 100% attribution proved in objective 2.

Budgeted but unfilled posts are identified and excluded from blended rates, since their cost is part-year and an hourly rate derived from it would be misleading.

Objective 5 — Performance layer Partial

Component	Records	Assessment
KPIs	43	Populated — 27 Customs, 16 Antiquities
Activity outputs	45	Populated — covering 45 activities
Maturity assessments	1	**Sparse** — one entity assessed

The model and the screens exist and hold real data for KPIs and outputs, which is what unit-cost reporting depends on. Maturity staging is implemented but only one assessment has been recorded, so cross-entity maturity comparison is not yet meaningful. This is a data-collection gap rather than a functional one.

Objective 6 — Governed budget cycle Not yet exercised

> **Reported as not proven**
> The submission and approval workflow is implemented, but there are currently
> **zero submissions**
> in the database. The cycle of prepare → submit → review → return → approve has not been run end to end against real departmental data, so no test result can honestly be claimed for it.

What *is* confirmed by inspecting the implementation: submissions are held per entity, cost centre and category; each category approves independently; budget lines lock once submitted; and a return does not edit the locked version but creates a new draft as the next version, retaining the prior one. A comparison report reads any two versions and reports what changed, alongside the dated record of who submitted, returned or approved and the notes they left.

This is the primary item recommended for the sandbox test window, where a full cycle can be run without touching live budget data. It is listed in the proposed test plan in section 7.

Objective 7 — Management reporting Passed

Reporting has been exercised in the application against the live data set. Available and working: income statement; general ledger view; project costs; activity costs on both a direct and a fully loaded basis; manpower allocation; employee cost per hour; entity budget summary; multi-year trend; a configurable report builder with cross-tabulation and charting; and an executive presentation populated with real figures.

All reports export to Excel and print in a layout designed for committee papers. A management review pack covers cost structure, capital discipline, cost per employee against a reference band, programme cost before and after allocation, and cost per output.

Objective 8 — Access control and auditability Passed

Control	Implementation confirmed
Password storage	PBKDF2-SHA256, 210,000 iterations, per-user salt — meets current OWASP guidance
Brute-force protection	Account lockout after 5 failed attempts; rate limiting on sign-in
Authorisation	Role-based, with a per-screen permission matrix enforced on every server request, not only in the interface
Data scoping	Users are confined to their entity and cost centre; scoping is applied in the data query
Audit trail	626 records, covering sign-in, imports, allocation runs, and data changes
Transport and browser	HTTPS enforced with HSTS; antiforgery protection; Content Security Policy permitting no external origin

Audit coverage observed: 344 sign-ins, 157 inserts, 80 updates, 17 allocation actions, 14 imports, plus recorded failed sign-in attempts.

Objective 9 — Operation without outbound internet Passed

Relevant to the Azure sandbox, where egress is expected to be restricted.

The entire application makes outbound network calls from exactly two places, both belonging to the optional AI assistant: the language model service, and the OECD statistics service. Both are disabled by a single configuration setting, and nothing else in the system depends on them.

All front-end libraries are served by the application itself rather than from public content delivery networks, and the Content Security Policy permits no external origin at all — so the browser is structurally unable to fetch a script, stylesheet or font from the internet. The application therefore runs fully functional in an air-gapped environment.

5. Known gaps and limitations

Stated in full so the evaluation is not based on an incomplete picture.

Gap	Detail and impact
Approval cycle unproven	Zero submissions exist; the workflow has never been run end to end with real data. Highest priority for sandbox testing.
Maturity data sparse	One assessment recorded. Cross-entity maturity comparison not yet meaningful.
Actuals partial	391 manpower actual records and 135 ledger actual records, covering part of the year and not all entities. Budget-versus-actual reporting is therefore complete only where actuals have been imported.
Pilot data incomplete	Two entities carry substantive budgets; a third holds three residual lines from earlier testing which should be removed before the sandbox is loaded.
Single-developer delivery	Built in-house without a second reviewer. Independent review during the sandbox phase would be prudent, and is welcomed.
Not yet load tested	Concurrency has not been tested beyond the current small user population of 9 accounts.

6. Proposed sandbox test plan

Once the Azure sandbox is provisioned, the following would close the open items above. The first is the most valuable, because it is the one thing that cannot be proven on the live system without disturbing a working budget.

1. **Full budget cycle, end to end.** Prepare a budget in a test entity, submit it, return it with comments, revise, resubmit, and approve through to final. Verify at each step that data locks and unlocks correctly, that a new version is created rather than the previous one overwritten, and that the comparison report and approval trail reproduce what happened.
2. **Concurrent multi-user test.** Several preparers working in different entities simultaneously, confirming data scoping holds under concurrent use.
3. **Permission matrix verification.** Confirm each role can reach exactly the screens intended and no others, tested against the server rather than the interface.
4. **Restricted-egress confirmation.** Run with outbound internet blocked and confirm every function except the optional assistant behaves normally.
5. **Restart and session persistence.** Recycle the application and confirm signed-in users remain signed in, proving the encryption key ring is correctly persisted.
6. **Backup and restore.** Take a backup, restore it, and confirm the recovery point and recovery time objectives already stated to EGA are achievable in the Azure environment.

The existing eight-step acceptance checklist already provided to EGA covers items 4 and 5 and can be run on the first day the environment is available.

7. Conclusion

The financial and costing objectives the system was built for are met and evidenced against real departmental data: manpower cost is fully attributed, overhead redistribution balances exactly, unit costing reconciles to total salary cost, and reporting is in daily use.

The performance layer is functional with real KPI and output data, though maturity assessment remains sparse. The approval workflow is built but unproven in practice, and proving it is the single most useful thing the sandbox can deliver.

We are ready to proceed with provisioning and to run the test plan in section 6 jointly with EGA.

GovBudget — Test Results and Business Objective Validation, 9 September 2026. Figures verified against the live database on the date of issue. Ras Al Khaimah Government, Department of Finance.